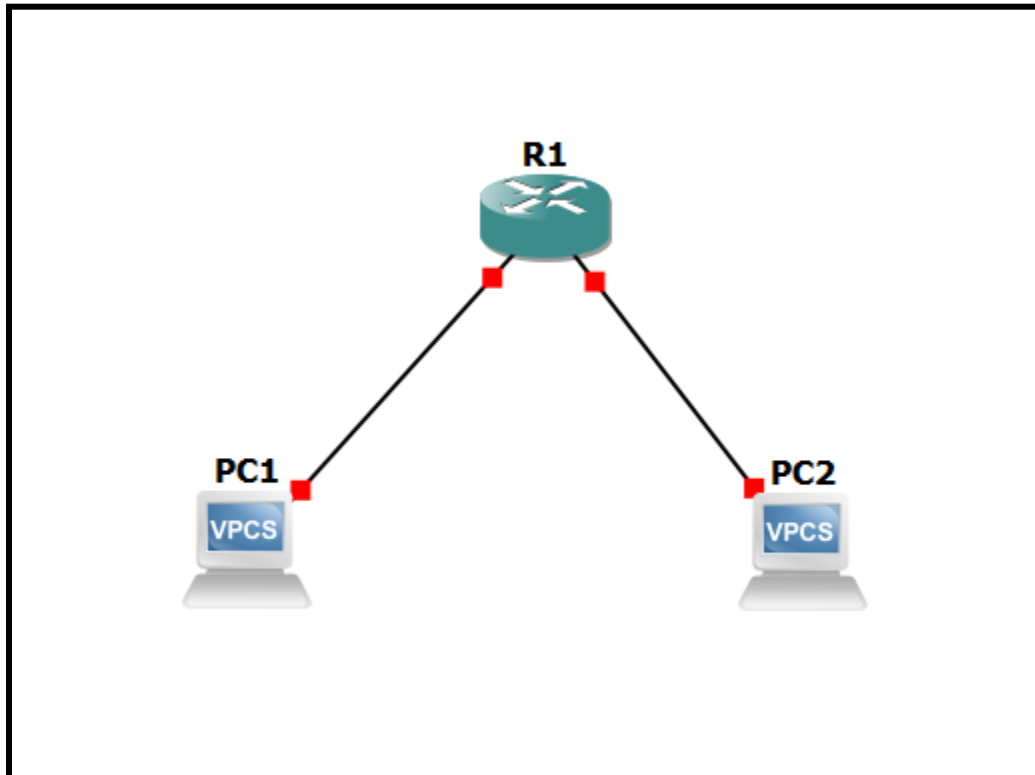


Part 1:-



Setting IP addresses to PC1 and PC2

```
PC1> ip 192.168.1.1 255.255.255.0  
Checking for duplicate address...  
PC1 : 192.168.1.1 255.255.255.0
```

```
PC2> ip 192.168.2.1 255.255.255.0  
Checking for duplicate address...  
PC1 : 192.168.2.1 255.255.255.0
```

Part 2:-

Pinging PC2 from PC1

→ The Pinging from PC1 to PC2 will not work as they are in 2 different networks

Router Configurations (Config)

```
R1#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)# interface fastEthernet 0/0
R1(config-if)#ip address 192.168.1.254 255.255.255.0
R1(config-if)#no shutdown
R1(config-if)#
*Mar  1 00:04:52.555: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Mar  1 00:04:53.555: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
```

→ This sets the ip address 192.168.1.254 to fastEthernet 0/0 with a mask of 255.255.255.0. So PC1 and fastEthernet 0/0 can communicate as they are in the same network. The IP address we set will be used as a gateway to communicate outside the network.

```
R1(config-if)# interface fastEthernet 0/1
R1(config-if)#ip address 192.168.2.254 255.255.255.0
R1(config-if)#no shutdown
R1(config-if)#
*Mar  1 00:05:45.215: %LINK-3-UPDOWN: Interface FastEthernet0/1, changed state to up
*Mar  1 00:05:46.215: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
```

→ This sets the ip address 192.168.2.254 to fastEthernet 0/1 with a mask of 255.255.255.0. So PC2 and fastEthernet 0/1 can communicate as they are in the same network. The IP address we set will be used as a gateway to communicate outside the network.

Pinging fastEthernet 0/0 and 0/1

```
PC1> ping 192.168.1.254
84 bytes from 192.168.1.254 icmp_seq=1 ttl=255 time=15.358 ms
84 bytes from 192.168.1.254 icmp_seq=2 ttl=255 time=15.783 ms
84 bytes from 192.168.1.254 icmp_seq=3 ttl=255 time=15.469 ms
84 bytes from 192.168.1.254 icmp_seq=4 ttl=255 time=15.524 ms
84 bytes from 192.168.1.254 icmp_seq=5 ttl=255 time=15.492 ms
```

→ PC1 pinging to fastEthernet 0/0

```
PC2> ping 192.168.2.254
84 bytes from 192.168.2.254 icmp_seq=1 ttl=255 time=15.227 ms
84 bytes from 192.168.2.254 icmp_seq=2 ttl=255 time=18.831 ms
84 bytes from 192.168.2.254 icmp_seq=3 ttl=255 time=15.102 ms
84 bytes from 192.168.2.254 icmp_seq=4 ttl=255 time=15.225 ms
84 bytes from 192.168.2.254 icmp_seq=5 ttl=255 time=15.270 ms
```

→ PC2 pinging to fastEthernet 0/1

→ Ping from PC1 to fastEthernet 0/1 did not work as they are located on different networks and hence don't know each other exists (we need to set the gateway for PC1 to use to communicate outside its own network)

Part 3:-

Setting the gateway for PC1 for full fledged communication

```
PC1> ip 192.168.1.1 192.168.1.254/24
Checking for duplicate address...
PC1 : 192.168.1.1 255.255.255.0 gateway 192.168.1.254
```

→ Setting the gateway for PC1 to use when communicating outside the network.

Pinging PC2 from PC1 but with set gateway

```
PC1> ping 192.168.2.1
192.168.2.1 icmp_seq=1 timeout
192.168.2.1 icmp_seq=2 timeout
192.168.2.1 icmp_seq=3 timeout
192.168.2.1 icmp_seq=4 timeout
192.168.2.1 icmp_seq=5 timeout
```

→ The ping request is being sent to PC2 from PC1 but PC2 is not able to respond as PC2's gateway is not set hence why it cant send back a reply across the network. Let's fix this!!

Setting the gateway for PC2 for full fledge communication

```
PC2> ip 192.168.2.1 192.168.2.254/24
Checking for duplicate address...
PC1 : 192.168.2.1 255.255.255.0 gateway 192.168.2.254
```

```
PC2> ping 192.168.1.1
84 bytes from 192.168.1.1 icmp_seq=1 ttl=63 time=30.932 ms
84 bytes from 192.168.1.1 icmp_seq=2 ttl=63 time=32.065 ms
84 bytes from 192.168.1.1 icmp_seq=3 ttl=63 time=30.192 ms
84 bytes from 192.168.1.1 icmp_seq=4 ttl=63 time=16.451 ms
84 bytes from 192.168.1.1 icmp_seq=5 ttl=63 time=30.823 ms
```

→ Once we set the gateway for PC2 to use for full fledge communication we can see that we get a reply back from PC1 when we ping from PC2!!.